AgentCompose — Supplementary Material

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This supplementary material supports the main paper *AgentCompose: Quantifying Compositional Generalization in LLM Tool Use* with extended tables, statistical test outputs, error analysis breakdowns, and tool / template inventories. Refer to the main paper for definitions and notation.

1 Per-Agent Per-Split Metric Tables

The full-metrics table in this supplement reports TSR, COC, and ABA for all five template agents across all four splits. The headline finding is that ChainAware uniquely achieves TSR = 1.000 on both *val* (in-distribution) and *test_substitutability* (with EquivAware); the other four agents plateau at TSR = 0.25 on substitutability because they never invoke the correct variant named in the natural-language prompt.

Agent	Split	N	TSR	COC	ABA
ReAct	val	400	0.268	0.125	0.057
	test_systematicity	600	0.428	0.060	0.055
	test_productivity	600	0.542	0.088	0.062
	test_substitutability	200	0.250	0.000	0.057
Native FC	val	400	0.268	0.125	0.284
	test_systematicity	600	0.145	0.060	0.1946
	test_productivity	600	0.200	0.000	0.196
	test_substitutability	200	0.250	0.000	0.297
Reflexion	val	400	0.268	0.125	0.057
	test_systematicity	600	0.428	0.060	0.055
	test_productivity	600	0.542	0.088	0.062
	test_substitutability	200	0.250	0.000	0.057
Plan-Execute	val	400	0.268	0.125	0.282
	test_systematicity	600	0.145	0.060	0.208
	test_productivity	600	0.107	0.000	0.213
	test_substitutability	200	0.250	0.000	0.292
ChainAware (Ours)	val	400	**1.000**	**1.000**	0.220
	test_systematicity	600	0.438	0.335	0.155
	test_productivity	600	0.452	0.000	0.178
	test_substitutability	200	**1.000**	0.205	0.221

Table 1: Per-agent per-split metric table (TSR, COC, ABA). ChainAware+EquivAware fully closes val and substitutability; other splits remain challenging across all agents.

2 DeepSeek-V4-Flash Validation

Bootstrap 95% confidence intervals (1000 iterations on per-task TSR) have half-widths ≤ 0.10 on every cell at N=100, confirming the qualitative pattern is statistically stable: even frontier 2026 LLMs hit near-zero TSR on productivity.

Agent	Val	Sys	Prod	Sub
DeepSeek-ReAct (N=200)	0.125	0.055	0.010	0.125
DeepSeek-FC (N=200)	0.155	0.070	0.020	0.160

Table 2: DeepSeek-V4-Flash per-split TSR (both agents at 200 instances per split, 1,600 tasks total). All cells complete; 95% bootstrap CIs are listed in the following table.

Agent / Split	TSR	95% CI	CGG	95% CI
DeepSeek-ReAct val	0.125	[0.080, 0.175]	—	—
DeepSeek-ReAct sys	0.055	[0.025, 0.090]	−0.070	[−0.125, −0.015]
DeepSeek-ReAct prod	0.010	[0.000, 0.025]	−0.115	[−0.165, −0.070]
DeepSeek-ReAct sub	0.125	[0.080, 0.175]	0.000	[−0.065, 0.060]
DeepSeek-FC val	0.155	[0.110, 0.205]	—	—
DeepSeek-FC sys	0.070	[0.040, 0.105]	−0.085	[−0.145, −0.025]
DeepSeek-FC prod	0.020	[0.005, 0.040]	−0.135	[−0.190, −0.080]
DeepSeek-FC sub	0.160	[0.110, 0.210]	0.005	[−0.065, 0.075]

Table 3: DeepSeek-V4-Flash TSR + CGG with 95% bootstrap confidence intervals (1000 iterations, seed 42; both agents at N=200 per split). CIs are reported in CGG only when statistically distinct from zero.

3 Statistical Test Outputs

3.1 Friedman Test

Split	χ^2	df	p-value
test_systematicity	41.7	4	< 0.001
test_productivity	87.4	4	< 0.001
test_substitutability	246.0	4	< 0.001

Table 4: Friedman test across 5 template agents, per-task paired ranks. Null hypothesis (all agents rank equally) rejected on every split.

3.2 Critical Difference

The CD diagram in the main paper provides the visual comparison. The CD threshold at $\alpha = 0.05$ for $k = 5$ algorithms and N tasks is 1.477 (Demsar 2006, Table 5). Agents whose mean ranks differ by more than 1.477 are significantly different.

3.3 Pairwise Wilcoxon (selected contrasts)

4 Error Analysis Breakdown

5 EquivAware Ablation

EquivAware is a deterministic, parameter-free overlay on ChainAware. We quantify its isolated effect on substitutability:

6 Tool Library (30 tools across 8 domains)

- **Information Retrieval** (5): search_web, search_news, fetch_url, summarize, translate
- **Computation** (5): calculator, unit_convert, statistics, round, sort

Split	Contrast	W	p_{uncorr}	Cohen's d
substitutability	ChainAware+EquivAware vs ChainAware	—	< 0.001	-2.44
substitutability	ChainAware+EquivAware vs Native FC	—	< 0.001	-2.44
productivity	ChainAware vs Native FC	—	< 0.001	-0.65
systematicity	ChainAware vs Native FC	—	< 0.001	-0.78

Table 5: Selected pairwise Wilcoxon signed-rank contrasts. Full 5×5 matrix in `results/statistical_tests.json` (attached as supplementary archive).

Error type	Val	Sys	Prod	Sub
missing_tool	0.73	0.86	0.80	1.00
wrong_tool	0.10	0.05	0.10	0.00
wrong_order	0.05	0.05	0.05	0.00
bad_argument	0.10	0.04	0.05	0.00

Table 6: Error-type breakdown for Native FC (rates among failed tasks). Missing-tool is the dominant failure mode across every split; 100% of substitutability failures are missing-tool.

- **Communication** (4): send_email, send_sms, send_push, send_slack
- **File I/O** (5): read_file, write_file, append_file, list_files, delete_file
- **Scheduling** (4): get_datetime, calendar_add, calendar_list, format_date
- **Data / Database** (4): sql_query, csv_filter, db_insert, json_transform
- **Image / Vision** (4): image_resize, image_crop, image_classify, ocr_extract
- **Audio / Speech** (3): audio_transcribe, text_to_speech, audio_classify

Six equivalence groups span these tools:

- G1_search: `search_web` ≡ `search_news`
- G2_communicate: `send_email` ≡ `send_sms`
- G3_data_query: `sql_query` ≡ `csv_filter`
- G4_visual_recognize: `image_classify` ≡ `ocr_extract`
- G5_notify: `send_slack` ≡ `send_push`
- G6_calendar_op: `calendar_add` ≡ `calendar_list`

Full JSON schemas for each tool are in `tools/*.json` (released with the code archive).

7 Task Templates (25 across 6 chain lengths)

The 25 task templates span chain lengths 2–5 and cross all 8 domains. Full YAML in `templates/task_templates.yaml`. Examples:

- *T1*: `search_web` → `send_email` — "Look up the weather in Paris and email it to a colleague."
- *T6*: `calculator` → `send_sms` — "Compute 42×17 and SMS the result to +1-555-0100."
- *T11*: `search_web` → `statistics` → `write_file` — "Fetch the last 30 days of AAPL closing prices, compute mean and stdev, and save as CSV."

Variant	Substitutability TSR	Cohen's d vs baseline
ChainAware (baseline)	0.250	—
ChainAware + EquivAware (Ours)	**1.000**	−2.44

Table 7: EquivAware lifts ChainAware's substitutability TSR from 0.250 (constant-tool baseline) to 1.000; no other split is affected.

- *T18*: `audio_transcribe` → `summarize` → `send_email` — "Transcribe `meeting.mp3`, summarize the action items, and email them to the team."

8 Chain-Length-Stratified TSR

Raw TSR is chain-length-confounded under the lenient evaluator: template agents call more tools on longer chains and are rewarded for it. The table below stratifies the seed-42 runs by expected chain length (n per cell in parentheses).

Split	Chain len	ReAct	Native FC	Plan-Execute	ChainAware
Sys.	3	0.21 (328)	0.21 (328)	0.21 (328)	**0.75** (328)
Sys.	4	0.61 (216)	0.00 (216)	0.00 (216)	0.00 (216)
Sys.	5	1.00 (56)	0.32 (56)	0.32 (56)	0.32 (56)
Prod.	4	**0.50** (553)	0.18 (553)	0.08 (553)	0.46 (553)
Prod.	5	1.00 (47)	0.40 (47)	0.40 (47)	0.40 (47)

Table 8: TSR by expected chain length (Reflexion omitted: identical to ReAct). ChainAware's matched-length advantage appears at 3-tool systematicity and in substitutability, not in 4-tool productivity, where ReAct's lenient call-everything behavior scores higher.

Interpretation: the in-sample "CGG" reading must control for chain depth; the paper's claims are limited to the matched-length comparisons highlighted above.

9 External Validity (Second Tool Library)

We constructed a second, disjoint tool library (20 tools: finance / iot / logistics, three equivalence groups; `tools_ext/`) with 12 task templates (`templates_ext/`), re-ran the five template agents plus ChainAware with library-agnostic inference (`-use-generic-inference`), and evaluated the same four splits at a reduced scale (the four split sizes are 100, 150, 150, and 50, respectively, from the generated sets).

Agent	Val	Systematicity	Productivity	Substitutability
ReAct / NativeFC / Reflexion / Plan-Execute	1.00	0.43	0.00	1.00
ChainAware	1.00	0.63	0.57	1.00

Table 9: TSR on the second tool library (generic inference). Every agent drops from the i.i.d. validation split (1.00) to systematicity (0.43) and productivity (0.00 for template agents; 0.57 for ChainAware), and ChainAware is best on both compositional axes. Full per-agent values and the verdict are in `results/external_validity.json`.

Verdict and limitations. This is *directional* external validity support: the qualitative claim that compositional splits are harder than i.i.d. holds on a second tool library, and ChainAware remains best on the compositional axes. Three limitations prevent a quantitative CGG-ordering comparison: (i) the external validation split is saturated (1.00 for all agents), so gap magnitudes are compressed; (ii) substitutability is also saturated (1.00 for all agents) because generic inference matches variant names in the natural-language

prompt and the equivalence-aware evaluator credits equivalent calls—the same mechanism that separates ChainAware on the main benchmark is here available to every agent; and (iii) the template-agent degeneracy (Reflexion≡ReAct, NativeFC≡PlanExecute) carries over to the external runs. The CGG-ordering Spearman correlations between main and external libraries are -0.745 (p=0.148) on systematicity and -0.725 (p=0.165) on productivity (n=5; bootstrap 95% CIs extend to -1.0); at $n = 5$ they are not statistically distinguishable from chance, so the external-validity claim rests on the systematic val-to-compositional drop (1.00 to 0.43/0.00), not on ordering correlation. The main benchmark's validation TSR (0.268) is unaffected by the external-validity driver bug that suppressed validation scores in an earlier external run: main results are produced by the original pipeline into `results/*_summary.json`, while external runs write only to `results/external/`.

10 Three-Way Ablation (Training × Detector × Scoring)

Seed-42 runs on the substitutability (200 instances) and productivity (600 instances) splits.

Cell	Substitutability TSR	Productivity TSR
ChainAware full (trained, detector on, equiv-aware scoring)	**1.00**	0.45
ChainAware no training	0.25	0.20
ChainAware no detector	1.00	0.45
ChainAware strict scoring	0.205	0.20
ReAct strict scoring	0.00	0.54

Table 10: Three-way ablation. Training is necessary for substitutability under equivalence-aware scoring ($0.25 \rightarrow 1.00$); the variant detector is inert on the metric (1.00 with or without it); and the equivalence-aware evaluator supplies the remaining credit—under strict scoring ChainAware drops to 0.205. On productivity, ReAct's lenient call-everything behavior scores higher than ChainAware under both scoring regimes.

11 Seed-Level Variance (5 independent seeds)

Template agents are deterministic given the task seed; across seeds 42, 1, 2, 3, 4 the task instances differ, yielding genuine variance. Ranges are min–max over the 5 seeds; full per-seed values in `results/seed_variance.json`.

Agent	Val	Systematicity	Productivity	Substitutability
ReAct	0.248 [0.2175, 0.268]	0.429 [0.412, 0.445]	0.513 [0.472, 0.542]	0.251 [0.210, 0.295]
Native FC	0.248 [0.2175, 0.268]	0.162 [0.145, 0.180]	0.184 [0.175, 0.200]	0.251 [0.210, 0.295]
Reflexion	0.248 [0.2175, 0.268]	0.429 [0.412, 0.445]	0.513 [0.472, 0.542]	0.251 [0.210, 0.295]
Plan-Execute	0.248 [0.2175, 0.268]	0.162 [0.145, 0.180]	0.104 [0.097, 0.112]	0.251 [0.210, 0.295]
Chain-Aware	**1.000**	0.476 [0.438, 0.520]	0.385 [0.293, 0.452]	**1.000**

Table 11: TSR mean and min–max range over 5 independent task seeds (split sizes are 400, 600, 600, and 200, respectively). Chain-Aware shows the largest productivity variance (0.293–0.452), so single-seed comparisons should use its 5-seed mean.

12 Hyperparameters and Configuration

- Random seed: 42 (default in all scripts)
- `max_steps`: 10 per task
- `n_synthetic_chains` (ChainAware): 50
- Bootstrap iterations: 1000

- Friedman / Wilcoxon α: 0.05 with Bonferroni correction
- DeepSeek temperature: 0 (deterministic)
- DeepSeek `max_tokens`: 1024 per turn

13 Bootstrap 95% Confidence Intervals (TSR)

1000 resamples, seed=42. Full per-agent values in `results/bootstrap_ci.json`.

Agent	Val	Sys.	Prod.	Sub.
ReAct	0.268 [0.225, 0.312]	0.428 [0.390, 0.467]	0.542 [0.503, 0.580]	0.250 [0.1949, 0.310]
Native FC	0.268 [0.225, 0.312]	0.145 [0.117, 0.173]	0.200 [0.168, 0.233]	0.250 [0.1949, 0.310]
Reflexion	0.268 [0.225, 0.312]	0.428 [0.390, 0.467]	0.542 [0.503, 0.580]	0.250 [0.1949, 0.310]
Plan-Execute	0.268 [0.225, 0.312]	0.145 [0.117, 0.173]	0.107 [0.083, 0.133]	0.250 [0.1949, 0.310]
ChainAware	1.000 [1.000, 1.000]	0.438 [0.400, 0.477]	0.452 [0.412, 0.492]	1.000 [1.000, 1.000]
DeepSeek-ReAct	0.125 [0.080, 0.175]	0.055 [0.025, 0.090]	0.010 [0.000, 0.025]	0.125 [0.080, 0.175]
DeepSeek-FC	0.155 [0.110, 0.205]	0.070 [0.040, 0.105]	0.020 [0.005, 0.040]	0.160 [0.110, 0.210]

Table 12: Task Success Rate (TSR) with bootstrap 95% CIs per split (both DeepSeek agents at N=200 per split).

Variance across independent training runs is not reported because all template agents are deterministic under the fixed seed 42; the bootstrap CIs above capture task-sampling variance. Chain-length-stratified success rates are deferred to the released raw per-task records (regenerable via `scripts/run_all.py -skip-llm`).

14 Reproducibility Checklist (NeurIPS-style)

- ✓ Code released (MIT license)
- ✓ Data released (3,000 tasks, 25 templates)
- ✓ Simulator released (deterministic JSON API)
- ✓ Random seeds documented
- ✓ Hyperparameters documented
- ✓ One-command reproduction (`python scripts/run_all.py -skip-llm`)
- ✓ Test suite (39/39 passing in <30 s)
- ✓ LLM API costs disclosed
- ✓ Statistical tests with effect sizes